

BAENGYEONGDO AQMS, Korea, Republic of

WMO: 471030

Lat: 37.966N

Lon: 124.630E

Elev: 413

StdP: 14.48

Time Zone: 9.00 (E09)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.2	19.2	8.9	8.8	20.0	11.6	10.1	21.3	29.3	28.3	25.8	28.1	10.3	30	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.5	84.6	77.1	82.3	75.7	80.3	74.5	79.0	82.7	77.5	80.6	76.2	78.9	5.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.8	146.8	81.0	76.8	141.7	79.6	75.2	134.3	78.1	42.9	84.3	41.2	79.7	39.7	78.9	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
24.3	20.2	17.1	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	12.1	80.7	2.8	2.0	10.1	82.1	8.4	83.3	6.8	84.5	4.8	85.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.7	29.7	31.8	38.8	48.8	58.1	66.9	73.4	75.3	68.4	58.4	45.9	36.1
	DBStd	16.58	6.13	6.22	5.00	6.03	5.02	4.36	4.17	3.88	4.19	5.62	7.36	6.29
	HDD50	2180	631	510	346	92	2	0	0	0	9	159	431	
	HDD65	5270	1096	930	811	488	221	30	0	17	209	573	895	
	CDD50	3182	0	0	0	55	252	507	725	784	553	270	36	0
	CDD65	797	0	0	0	1	6	87	260	319	119	5	0	0
	CDH74	3205	0	0	0	3	15	188	1095	1689	212	3	0	0
	CDH80	533	0	0	0	1	0	18	207	300	7	0	0	0
Wind	WSAvg	7.6	8.8	7.6	8.8	9.1	8.1	6.5	6.8	6.5	6.0	6.7	8.3	8.3
Precipitation	PrecAvg	32.2	0.5	0.7	0.7	2.0	2.8	3.0	8.6	6.5	3.4	1.4	1.6	0.9
	PrecMax	49.8	1.1	2.6	2.8	3.9	5.4	8.0	15.7	12.9	14.3	3.7	6.1	1.7
	PrecMin	16.0	0.1	0.0	0.1	0.2	0.1	0.6	2.9	1.4	0.2	0.0	0.4	0.3
	PrecStd	7.9	0.3	0.6	0.7	1.1	1.7	1.8	3.7	3.4	4.1	1.1	1.5	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.1	46.8	53.8	70.3	75.6	82.4	88.2	87.9	80.9	72.0	62.0	52.0
		MCWB	43.5	43.4	46.0	57.8	65.2	67.6	79.5	78.2	74.0	65.6	57.0	48.3
	2%	DB	42.8	44.3	50.1	64.4	71.3	78.5	84.6	85.7	78.7	69.6	59.3	49.0
		MCWB	40.6	41.1	44.4	54.7	60.9	67.2	77.6	77.8	72.8	63.6	54.4	46.2
	5%	DB	41.0	41.4	47.9	60.7	68.1	75.6	82.1	82.8	76.8	67.5	57.3	46.7
		MCWB	38.6	38.4	43.1	52.1	58.7	66.9	75.7	76.6	71.1	61.7	53.1	43.6
	10%	DB	38.9	39.5	45.8	57.4	65.9	73.5	79.2	80.9	74.9	64.8	55.4	44.8
		MCWB	36.3	36.7	41.4	49.8	57.6	65.4	74.6	75.4	69.6	59.7	51.5	41.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.9	45.6	47.7	61.3	65.6	72.0	81.3	81.2	77.3	67.4	59.3	49.8
		MCDB	45.1	46.6	50.4	67.1	73.5	76.2	85.8	85.0	79.2	70.6	60.8	51.0
	2%	WB	40.9	41.4	45.5	55.8	62.7	70.2	78.9	79.3	74.7	64.7	56.6	46.7
		MCDB	42.1	42.9	48.7	62.2	68.8	74.0	82.8	83.0	76.8	67.8	58.3	48.3
	5%	WB	38.9	39.4	43.6	53.4	60.6	69.1	77.4	78.1	72.4	62.9	54.0	44.3
		MCDB	40.5	41.0	46.7	59.2	65.5	72.6	80.4	81.5	75.4	66.0	56.4	46.3
	10%	WB	35.9	37.2	41.7	51.1	59.0	68.0	76.0	76.8	70.3	61.1	52.1	41.9
		MCDB	38.2	39.4	44.4	55.9	63.9	71.1	78.4	79.8	73.9	64.5	54.9	44.5
Mean Daily Temperature Range	5% DB	MDBR	9.4	9.2	10.0	11.9	11.8	10.3	8.4	9.5	9.8	9.4	9.2	8.8
		MCDBR	11.2	10.0	12.4	16.5	15.1	13.8	12.0	11.6	11.2	10.7	10.3	10.1
		MCWBR	10.6	8.6	9.5	11.0	10.0	8.6	8.6	7.2	7.7	8.4	9.3	9.9
	5% WB	MCDBR	11.5	9.0	11.5	16.3	13.1	10.8	10.5	10.2	10.2	10.1	10.4	10.0
		MCWBR	11.6	8.7	9.6	11.8	9.8	7.7	8.0	7.0	7.9	9.3	10.8	10.3
Clear-Sky Solar Irradiance	taub	0.432	0.485	0.543	0.594	0.613	0.693	0.670	0.576	0.505	0.517	0.498	0.439	
	taud	1.961	1.860	1.760	1.702	1.716	1.627	1.725	1.941	2.038	1.883	1.878	1.948	
	Ebn at Noon	221	226	226	225	224	207	210	227	236	213	196	207	
	Edh at Noon	43	54	66	74	74	81	73	58	49	52	47	41	
All-Sky Solar Radiation	RadAvg	698	995	1388	1601	1759	1619	1392	1499	1419	1131	749	605	
	RadStd	54	74	115	110	163	197	187	181	121	78	88	43	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+151	+86

Nomenclature: See separate page